

Sims Framework Contract

Scope

This framework is fully independent from `mut_var` internals and writes outputs only under: - `sims/results/`
- `sims/plots/`

It must remain compatible with `mut_var` inference by file contract.

Methods-doc readability requirement

Any methods/design Markdown in `sims/docs/` must also be rendered to PDF for human review.

Canonical command:

```
python sims/scripts/render_methods_pdf.py --docs-dir sims/docs --out-dir sims/docs/docs_pdf
```

This keeps generated human-readable artifacts within `sims/docs/`.

Canonical `mut_var` entrypoints

- Inference: `mutvar infer <observed.tsv> -o <infer.tsv>`
- Curve fit: `mutvar curve <infer.tsv> --fit-only -o <curve.tsv>`

No changes are made to `src/`, `tests/`, packaging, or dependencies.

Required observed schema

Observed TSV must include these exact columns: 1. `effect_allele_frequency` 2. `beta` 3. `standard_error`

Additional columns are allowed but optional.

Core simulation controls

The simulator must expose and record: - `frequency.demography_mode` (currently `equilibrium` implemented) - `effect.selection_mode` in `{hd, ld}` - `ascertainment.ascertainment_statistic` with primary default `threshold_on_maf` and optional alternative modes `{threshold_on_v_true, threshold_on_v_hat}` - `ascertainment.maf_min` for MAF-threshold runs - `ascertainment.v_s_cutoff` and `ascertainment.p_value_threshold` for variance-threshold runs - `dfe.point_mass_zero` in `[0, 1)` with default `0.0`, defining the DFE atom at `beta_s = 0`; the tabulated SSD grid parameterizes the nonzero DFE component - `effect.trait_null_fraction` in `[0, 1]` with default `0.0`, defining an additional focal-trait null gate on top of latent DFE draws - fixed `n_ascertained` target per run - `truth_reference_n` for distribution-evaluation truth sampling (should be larger than `n_ascertained`)

Runtime caution: variance-threshold scenarios are intentionally retained for comparison checks but can be extremely sensitive to `v_s_cutoff` and may run for a long time if configured too strictly. For routine workloads, `threshold_on_maf` is the operational default in this framework.

Single-threshold contract for MAF ascertainment: when `ascertainment.ascertainment_statistic = threshold_on_maf`, generation support and ascertainment use the same threshold (`frequency.min_x` is overridden by `ascertainment.maf_min` for that run). For v-based ascertainment modes, generation support remains controlled by `frequency.min_x`.

The observed `beta` column is on the selection-scaled effect axis (`beta_s`), and metadata must record the canonical scaled coefficient definition $S_{ud} = 2N_e s_{ud}$.

When `dfe.point_mass_zero > 0`, the DFE includes a nonzero atom at `beta_s = 0` in latent effects before noise and ascertainment. Observed outputs are not required to preserve an exact zero fraction.

When `effect.trait_null_fraction > 0`, additional loci can have latent `beta_s=0` even when selected/neutral draws remain nonzero under the DFE. Observed outputs are not required to preserve an exact zero fraction.

Domain and quality constraints

For every row in observed TSV: - `effect_allele_frequency` is finite and in $[0, 1]$ - `beta` is finite - `standard_error` is finite and strictly > 0 - No nulls in required columns - File is tab-separated with a header row

Simulator output artifacts per run

Given `run_id`, simulator writes: - `sims/results/<run_id>.observed.tsv` (`mut_var` input) - `sims/results/<run_id>.truth.tsv` (latent variables and generating params, including zero-source flags; may use an independent larger truth-reference sample) - `sims/results/<run_id>.meta.tsv` (run metadata and summary stats) - `sims/results/<run_id>.manifest.json` (reproducibility metadata)

Downstream `mut_var` outputs: - `sims/results/<run_id>.infer.tsv` - `sims/results/<run_id>.curve.tsv`

Recovery objective

Primary study question: - How well `mut_var`'s inferred mixture captures the underlying DFE-generated effect-size distribution.

Evaluation compares: - Truth-side unconditional effect distribution summaries - `mut_var` inferred mixture summaries reconstructed from `infer.tsv` (`mu0`, `var0`, and mixture weights) - Distribution distances and tail-fidelity metrics on $|\beta_s|$

AF-conditioned diagnostics may be computed as optional secondary outputs, but they are not the primary criterion in this workflow.

Reproducibility requirements

Each run must be reproducible from: - random seed - scenario configuration - exact commands executed - artifact paths and basic checks (row counts and key summary values)

Simulator-side zero-gate contracts are validated under `sims/` with:

```
python sims/scripts/check_zero_gate_contract.py
```